PROFINITE RECOGNITION OF MAHLER EQUATIONS AND D-FINITE WITT ESCAPE

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ABSTRACT. Fix an integer base $k \geq 2$ and a power series $F(t)$ with rational coefficients. We prove that Mahler relations in a fixed order–degree box are recognized by arbitrarily accurate prime-power reductions, even when the modular relations are mutually incompatible. At a fixed prime the exact maximal false-positive depth is the stabilized largest Smith exponent of a sufficiently large full-rank coefficient truncation. Across primes, one nonzero minor gives a logarithmic precision budget, and divergence of the total local precision is equivalent to an exact relation.

We also prove a nonlinear recognition theorem: if, to unbounded p-adic depth, a bounded-degree multiplier makes the full Cartier matrix have bounded determinantal rank, then F is a characteristic-zero Mahler function. Combining these mechanisms with classical theorems gives three rigidity consequences. Persistence in two multiplicatively independent bases is equivalent to rationality; every nonrational D-finite series escapes each fixed Witt–Mahler box; and its reduced p-kernel cardinalities tend to infinity. When the reductions are automatic, the last statement is an automaton-state lower conclusion.

An adelic one-pole hypothesis gives a further unconditional consequence: the reciprocal of the unique pole in the complex open unit disk is a Pisot or Salem number. Finally, sharp boundary examples separate fixed-base depth from changing-characteristic phenomena. The central-binomial series has full own-p Cartier rank one modulo p, while the least coefficient degree of an endpoint-nonzero own-p relation is exactly $(p-1)/2$. We also exhibit a series rational modulo every p^m but non-Mahler in every integer base. No effective escape rate is claimed.

1. INTRODUCTION

A nonzero formal power series F is called k-Mahler if it satisfies an identity

$$\sum_{i=0}^{q} P_i(t)F(t^{k^i}) = 0, \qquad P_i(t) \in \mathbf{Q}[t], \qquad P_0P_q \neq 0. \tag{1}$$

We regard the zero series as Mahler in every base. A relation initially missing its $F(t)$ endpoint can be normalized to this form; the elementary residue-section argument is recorded in Lemma 2.1 below. Reduction modulo p^m often produces identities of this shape long before one knows whether an identity exists in characteristic zero. The main question is deliberately

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uniform: what follows if the base, order and coefficient degrees remain bounded while arithmetic precision tends to infinity, vertically at one prime, horizontally through many primes, or by a mixture of the two?

The answer is exact. Compactness gives a p-adic relation without requiring the modular coefficient vectors to be compatible, and a finite-minor rank argument descends it to $\mathbf{Q}$. Smith normal form then identifies the last possible false-positive depth. This linear mechanism has a nonlinear counterpart in which a bounded polynomial "blocks" the poles before a uniform Cartier-rank constraint is imposed.

We use *Witt depth* only as shorthand for passage through the coefficient rings $\mathbf{Z}/p^m\mathbf{Z}$. No operation from the formal theory of Witt vectors is used.

One consequence no longer requires D-finiteness. Two exact Mahler relations in multiplicatively independent bases force rationality by the theorem of Adamczewski and Bell [2]; our recognition theorem turns this into an if-and-only-if criterion stated solely in terms of prime-power persistence. The principal escape consequences concern D-finite series. Bézivin's theorem, in the formulation of Bell, Coons and Rowland [5], says that a series which is both D-finite and k-Mahler is rational. Hence modular Mahler relations for a nonrational D-finite series cannot remain in a fixed box through all Witt depths. This is a complexity escape statement, not a quantitative lower bound.

Our results are organized as follows. Section 2 gives fixed-box profinite recognition and the endpoint normalization needed by later applications. Section 3 computes exact local precision, proves mixed-prime recognition, and derives the two-base rationality criterion. Section 4 proves the multiplier–Cartier recognition theorem. Sections 5 and 6 derive D-finite and automata consequences. Section 7 treats the changing-base obstruction, Section 8 gives sharpness examples, and Section 9 proves the adelic pole reversal.

The arguments combine elementary compactness and determinantal algebra with classical inputs of Becker [3], Bézivin [4], Adamczewski–Bell [2], Adamczewski–Bell–Smertnig [1], and Rowland–Yassawi [7, 9]. Our contribution is the exact fixed-box threshold, the incompatible-blocker recognition principle, and the escape and profinite reformulations obtained by assembling these tools. The two-base rationality implication and the Mahler-denominator input in Section 9 are explicitly credited corollaries of the cited theorems. We do not claim a solution of Christol's conjecture, a general diagonal conjecture, or the Salem–Boyd closure problem.

2. Fixed-box profinite recognition

Throughout, $k \geq 2$ and p is prime. Let

$$F(t) = \sum_{n \geq 0} a_n t^n \in \mathbf{Q}[[t]] \cap \mathbf{Z}_p[[t]].$$

For $q \geq 1$ and $h \geq 0$ define the finite family

$$\mathcal{V}_{q,h}^{(k)}(F) = \{t^j F(t^{k^i}) : 0 \leq i \leq q,\ 0 \leq j \leq h\}. \tag{2}$$

When the base is fixed we write simply $\mathcal{V}_{q,h}(F)$. It has $N = (q+1)(h+1)$ elements. A coefficient vector in $(\mathbf{Z}/p^m\mathbf{Z})^N$ is called *primitive* if at least one coordinate is a unit modulo p.

The next elementary point prevents a hidden endpoint gap when a linear dependence is converted into a standard Mahler equation.

Lemma 2.1 (Endpoint normalization)**.** *Let K be a field, $F \in K[[t]]$ be nonzero, and suppose*

$$\sum_{i=s}^{q} P_i(t) F(t^{k^i}) = 0, \qquad P_s \neq 0, \tag{3}$$

with $P_i \in K[t]$. Then F satisfies a k-Mahler equation whose coefficient of $F(t)$ is nonzero.

Proof. Write, uniquely,

$$P_i(t) = \sum_{r=0}^{k^s-1} t^r P_{i,r}(t^{k^s}).$$

The residue classes of exponents modulo k^s in (3) do not interact. On putting $u = t^{k^s}$, each residue class therefore gives

$$\sum_{i=s}^{q} P_{i,r}(u) F(u^{k^{i-s}}) = 0.$$

Choose r with $P_{s,r} \neq 0$ and delete any zero terms at the far end. The resulting relation begins with a nonzero multiple of $F(u)$. It must also contain a later term, since otherwise $P_{s,r}F = 0$, contrary to the hypotheses. □

Theorem 2.2 (Vertical profinite recognition)**.** *For fixed k, q, h, the following are equivalent.*

(i) *The family $\mathcal{V}_{q,h}(F)$ is linearly dependent over $\mathbf{Q}$.*
(ii) *For every $m \geq 1$ there is a primitive coefficientwise relation among the family modulo p^m.*
(iii) *Such a primitive relation exists for arbitrarily large m.*

No compatibility between the relations at different depths is required.

Proof. The implication (i)⇒(ii) follows after multiplying a rational relation by a scalar that makes its coefficients p-integral and primitive. The implication (ii)⇒(iii) is immediate.

For (iii)⇒(i), lift the modular coefficient vectors to the compact p-adic unit shell

$$S = \{c \in \mathbf{Z}_p^N : \max_j |c_j|_p = 1\}.$$

Along depths tending to infinity, take a convergent subsequence with limit $c \in S$. Each coefficient of the formal series

$$\sum_{i=0}^{q} \sum_{j=0}^{h} c_{i,j} t^j F(t^{k^i})$$

is a continuous $\mathbf{Z}_p$-linear function of c. Its values along the subsequence lie in ideals $p^m \mathbf{Z}_p$ with $m \to \infty$, so its value at the limit is zero. This gives a nonzero dependence over $\mathbf{Q}_p$.

The coefficient matrix has rational entries. If its columns were independent over $\mathbf{Q}$, some finite rational N-by-N minor would be nonzero, and it would remain nonzero over $\mathbf{Q}_p$. Thus the dependence descends to $\mathbf{Q}$. □

For $F \neq 0$, every exact dependence furnished by Theorem 2.2 becomes a standard k-Mahler equation by Lemma 2.1. The zero series is already Mahler by convention.

Remark 2.3. Fixed base, fixed q, h, and primitivity are essential. Allowing the box to grow with m, or allowing all coefficients of a relation to be divisible by p, destroys the conclusion. Section 7 explains a different phenomenon when the base itself changes with p.

3. Exact precision and finite certificates

Let $M = M_{q,h}(F)$ be the infinite coefficient matrix whose N columns are the series in (2). First suppose that M has column rank N. For a finite row set R on which M_R has rank N, view M_R over $\mathbf{Z}_p$ and write its Smith invariant factors as

$$d_1(R) \mid \cdots \mid d_N(R), \qquad \alpha_R = v_p(d_N(R)).$$

Set

$$\epsilon_p(M) = \sup\big(\{m \geq 1 : Mx \equiv 0 \pmod{p^m} \text{ has a primitive solution}\} \cup \{0\}\big). \tag{4}$$

Proposition 3.1 (Exact local depth)**.** *The exact maximal primitive depth of the full infinite system is*

$$\epsilon_p(M) = \min_R \alpha_R, \tag{5}$$

where R ranges over finite full-rank row sets. Along any increasing full-rank exhaustion of the rows, the corresponding exponents decrease and eventually stabilize at $\epsilon_p(M)$.

Proof. Unimodular row and column operations preserve primitivity. For a finite matrix with Smith diagonal $d_1, \ldots, d_N$, a primitive kernel vector modulo p^m exists exactly when $m \leq v_p(d_N)$. This proves the upper bound in (5).

Conversely, suppose $m \leq \alpha_R$ for every finite full-rank R. Every finite set of row equations can be enlarged to a full-rank one and hence has a primitive solution modulo p^m. The finite set of primitive vectors in $(\mathbf{Z}/p^m\mathbf{Z})^N$

therefore has the finite-intersection property for these row constraints, so one vector solves all rows. Adding rows can only decrease the exponent. A nonincreasing sequence of nonnegative integers stabilizes, which proves the final assertion. □

If M has rank less than N, there is an exact relation and the available depth is infinite. Proposition 3.1 concerns precisely the full-rank case.

The same argument provides a global obstruction. Let $\Delta_0 \neq 0$ be a fixed rational N-by-N minor of M. Write it in lowest terms.

Proposition 3.2 (Single-minor precision budget)**.** *Let $\mathcal{P}$ be a finite set of primes at which all entries used in Δ_0 are integral. If a primitive relation exists modulo p^{m_p} for every $p \in \mathcal{P}$, then*

$$\sum_{p\in\mathcal{P}} m_p \log p \le \log|\operatorname{num}(\Delta_0)|. \tag{6}$$

In particular, a nonzero minor is an explicit certificate that relations in this box cannot persist past its local valuations.

Proof. If $Bx \equiv 0 \pmod{p^{m_p}}$ for the square minor matrix B and x is primitive, the adjugate identity implies $v_p(\det B) \ge m_p$. Hence p^{m_p} divides the numerator of Δ_0. Multiplication over $p \in \mathcal{P}$ and taking logarithms gives (6). □

There is also an integral finite-truncation version. For a finite full-rank M_R, let D_R clear its denominators and let $s_N(R)$ be the largest integer Smith invariant of $A_R = D_R M_R$. At primes $p \nmid D_R$,

$$m_p \le v_p(s_N(R)), \qquad \sum_{p\in\mathcal{P}} m_p \log p \le \log|s_N(R)|. \tag{7}$$

We emphasize the direction: a nonzero minor or Smith invariant gives a *negative* certificate against deeper lifting. Without an independent row cutoff it is not a finite positive certificate for the existence of an exact relation.

3.1. **Vertical, horizontal and mixed precision.** We now reset the ambient arithmetic. For $F = \sum a_n t^n \in \mathbf{Q}[[t]]$, call a prime p *good for* F if $a_n \in \mathbf{Z}_p$ for every n. At a good prime, let $\epsilon_p^{(b;q,h)}(F) \in \mathbf{N} \cup \{0, +\infty\}$ be the supremum of 0 and all depths $m \ge 1$ at which $\mathcal{V}_{q,h}^{(b)}(F)$ has a primitive coefficientwise relation modulo p^m.

Theorem 3.3 (Mixed vertical–horizontal recognition)**.** *Let $b \ge 2$, $q \ge 1$, $h \ge 0$, and let $\mathcal{P}$ be a nonempty set of primes good for $F \in \mathbf{Q}[[t]]$. Then*

$$\mathcal{V}_{q,h}^{(b)}(F) \text{ is dependent over } \mathbf{Q} \iff \sum_{p\in\mathcal{P}} \epsilon_p^{(b;q,h)}(F)\log p = +\infty. \tag{8}$$

If the family is independent and $\Delta_0 \neq 0$ is any rational maximal minor of its coefficient matrix, then more precisely

$$\sum_{p\in\mathcal{P}} \epsilon_p^{(b;q,h)}(F)\log p \le \log|\operatorname{num}(\Delta_0)|. \tag{9}$$

A summand with value $+\infty$ makes the sum infinite. For an infinite prime set, the sum means the supremum of its finite sub-sums.

Proof. An exact rational relation can be cleared to a primitive integer vector. It is primitive at every prime and gives relations through every depth, proving the forward implication. Conversely, suppose the family is independent and choose Δ_0. For every finite $\mathcal{P}_0 \subset \{p \in \mathcal{P} : \epsilon_p^{(b;q,h)}(F) \geq 1\}$, Proposition 3.2, applied at the maximal finite available depths, gives

$$\sum_{p \in \mathcal{P}_0} \epsilon_p^{(b;q,h)}(F) \log p \leq \log |\operatorname{num}(\Delta_0)|.$$

If one local depth is infinite, apply the same argument at arbitrarily large finite depths. Taking the supremum over finite $\mathcal{P}_0$ proves both (9) and the reverse implication. □

Thus one fixed prime of unbounded depth is the *vertical* case; infinitely many good primes of positive depth give the *horizontal* case; and every divergent weighted mixture is allowed. The relations may vary with both prime and depth, but the base and the box may not vary.

Theorem 3.4 (Two-base profinite rationality)**.** *Let $k, \ell \geq 2$ be multiplicatively independent and let $F \in \mathbf{Q}[[t]]$. The following are equivalent.*

(i) *$F \in \mathbf{Q}(t)$.*
(ii) *For each $b \in \{k, \ell\}$ there are fixed $q_b \geq 1$, $h_b \geq 0$ and a nonempty set $\mathcal{P}_b$ of primes good for F such that*

$$\sum_{p \in \mathcal{P}_b} \epsilon_p^{(b;q_b,h_b)}(F) \log p = +\infty.$$

The two prime sets, depths and modular relations need not agree.

Proof. Assume (ii). Theorem 3.3, followed by Lemma 2.1 when $F \neq 0$, makes F both k-Mahler and ℓ-Mahler. The theorem of Adamczewski and Bell [2, Theorem 1.3] gives $F \in \mathbf{Q}(t)$.

Conversely, let $F = A/B \neq 0$ with $A, B \in \mathbf{Z}[t]$ and $B(0) \neq 0$. For every $b \geq 2$,

$$A(t^b)B(t)F(t) - A(t)B(t^b)F(t^b) = 0. \tag{10}$$

After division by the joint integer content, its coefficient vector is primitive at every prime. Every $p \nmid B(0)$ is good for F, and (10) persists through every depth at any such prime. This proves (ii). For $F = 0$, use the relation $F(t) = 0$. □

Theorem 3.4 is a profinite reformulation obtained by combining Theorem 3.3 with the characteristic-zero two-base theorem of Adamczewski and Bell; no new proof of their rationality theorem is claimed here.

4. Polynomial blockers and Cartier rank

For $0 \le r < k$ define the Cartier section

$$\Lambda_{k,r}\left(\sum_{n\ge 0} b_n t^n\right) = \sum_{n\ge 0} b_{kn+r} t^n.$$

For a word w in the digits $0,\dots,k-1$, let Λ_w be the corresponding iterate. The full Cartier matrix of a series G is

$$\mathcal{C}_k(G) = \left([t^n]\Lambda_w G\right)_{n\ge 0,\ w}. \tag{11}$$

Its column rank is the dimension of the linear span of the k-kernel of G.

Proposition 4.1 (Raw Cartier rank cannot preserve an inner pole)**.** *If $G(t) = \sum_{n\ge 0} b_n t^n \in \mathbf{C}[[t]]$ has finite-dimensional full k-Cartier span, then $|b_n| = O(n^c)$ for some $c \ge 0$. In particular, the complex radius of convergence of G is at least 1.*

Proof. On the finite-dimensional span of the Cartier sections, the k section operators are represented by finitely many matrices. Reading the base-k digits of n expresses b_n as a fixed linear functional applied to a product of $O(\log n)$ of these matrices. A common operator-norm bound therefore gives $|b_n| \le C A^{O(\log n)} = O(n^c)$. □

Thus finite raw Cartier rank cannot hold for a series with a pole strictly inside the complex unit disk. The multiplier below is not cosmetic: it may cancel finitely many poles before rank is measured.

Definition 4.2. A polynomial $Q(t) = \sum_{j=0}^{M} q_j t^j$ over $\mathbf{Z}/p^m\mathbf{Z}$ is a *primitive (M, D) blocker for F at depth m* if some q_j is a unit modulo p and every $(D+1)$-minor of $\mathcal{C}_k(QF)$ vanishes modulo p^m.

Theorem 4.3 (Nonlinear profinite recognition)**.** *Let p be prime, let $F \in \mathbf{Q}[[t]] \cap \mathbf{Z}_p[[t]]$, and fix $k \ge 2$ and $M, D \ge 0$. If primitive (M, D) blockers exist for arbitrarily large m, then F is k-Mahler over $\mathbf{Q}$.*

Proof. Lift blocker coefficients to the compact unit shell in $\mathbf{Z}_p^{M+1}$. For each m, let X_m be the closed subset on which every indicated minor lies in $p^m\mathbf{Z}_p$. The hypothesis makes X_m nonempty and $X_{m+1} \subseteq X_m$. Compactness gives a point $Q \in \bigcap_m X_m$. It is nonzero, has degree at most M, and every $(D+1)$-minor of $\mathcal{C}_k(QF)$ vanishes exactly. Thus $G = QF$ has a finite-dimensional k-kernel over $\mathbf{Q}_p$.

Finite Cartier span means that G is k-regular. Becker's theorem [3, Theorem 1] then gives a k-Mahler equation over $\mathbf{Q}_p(t)$. Substituting $G(t^{k^i}) = Q(t^{k^i})F(t^{k^i})$ and clearing denominators gives some finite relation among the iterates of F over $\mathbf{Q}_p[t]$. The corresponding finite coefficient matrix has rational entries, because $F \in \mathbf{Q}[[t]]$. The same finite-minor rank descent used in Theorem 2.2 produces a relation over $\mathbf{Q}[t]$; if $F \ne 0$, Lemma 2.1 puts it in standard Mahler form. □

Remark 4.4. The limiting blocker need not have rational coefficients. The proof gives existence of a characteristic-zero Mahler equation but, without an effective form of the Becker and elimination steps, gives no bound for its eventual order or coefficient degree in terms of k, M, D.

The nonlinear system also has a horizontal and mixed-prime obstruction, although its certificate is a projective Nullstellensatz identity rather than one determinant. At a good prime, let $\eta_p^{(b;M,D)}(F) \in \mathbf{N} \cup \{0, +\infty\}$ be the supremum of 0 and all depths $m \geq 1$ at which a primitive (M, D) blocker in base b exists modulo p^m.

Theorem 4.5 (Mixed-prime blocker recognition)**.** *Fix $b \geq 2$ and $M, D \geq 0$. If $F \in \mathbf{Q}[[t]]$ is not b-Mahler, then there is a nonzero integer $A = A(F; b, M, D)$ such that, for every set $\mathcal{P}$ of good primes,*

$$\sum_{p \in \mathcal{P}} \eta_p^{(b;M,D)}(F) \log p \leq \log |A|. \tag{12}$$

Consequently, divergence of the sum forces F to be b-Mahler.

Proof. Regard every $(D+1)$-minor of $\mathcal{C}_b(QF)$ as a homogeneous polynomial $f_\nu(q_0, \ldots, q_M)$ over $\mathbf{Q}$. If all these polynomials had a common nonzero zero over $\overline{\mathbf{Q}}$, the corresponding algebraic polynomial Q would make QF have finite Cartier rank. Becker's theorem, substitution, rational-column rank descent and Lemma 2.1 would then make F b-Mahler. Under the stated hypothesis the projective common zero set is therefore empty.

By Hilbert's basis theorem, finitely many of the f_ν generate the same homogeneous ideal. The projective Nullstellensatz puts a power of each q_j in its scalar extension to $\overline{\mathbf{Q}}$; faithful flatness descends these ideal-membership certificates to $\mathbf{Q}$. After clearing denominators, there are an integer $A \neq 0$, an exponent L, and integer polynomials $H_{j,\nu}$ such that

$$A q_j^L = \sum_\nu H_{j,\nu}(q) g_\nu(q) \qquad (0 \leq j \leq M), \tag{13}$$

where each g_ν is an integer multiple of the corresponding minor. For a primitive blocker modulo p^m, choose a lift $\widetilde{q} \in \mathbf{Z}_p^{M+1}$. Goodness of p makes the selected minor polynomials p-integral, some $\widetilde{q}_j$ is a unit, and all $g_\nu(\widetilde{q})$ vanish modulo p^m; hence $p^m \mid A$. Thus $\eta_p^{(b;M,D)}(F) \leq v_p(A)$ at every good prime. Summing over finite prime sets and taking the supremum proves (12). □

The algebraic blocker arising in the first paragraph need not have rational coefficients. What descends to $\mathbf{Q}$ is the resulting Mahler equation for the rational-coefficient series F.

Corollary 4.6 (Two-base blocker criterion)**.** *Let $k, \ell \geq 2$ be multiplicatively independent and $F \in \mathbf{Q}[[t]]$. Then F is rational if and only if, for each*

$b \in \{k, \ell\}$, there are fixed M_b, D_b, a nonempty set $\mathcal{P}_b$ of good primes, and

$$\sum_{p \in \mathcal{P}_b} \eta_p^{(b;M_b,D_b)}(F) \log p = +\infty.$$

Proof. Assume the blocker condition. Theorem 4.5 makes F Mahler in both bases, so Adamczewski–Bell gives $F \in \mathbf{Q}(t)$. Conversely, write $F = A/B$ with $A, B \in \mathbf{Z}[t]$ and $B(0) \neq 0$. At every $p \nmid B(0)$, the primitive multiplier $Q = B$ makes $QF = A$ a polynomial; all its Cartier sections lie in a space of dimension at most $\deg A + 1$. Thus fixed bounds work at every depth. The zero series is immediate with $Q = 1$. □

Corollary 4.7 (Archimedean obstruction to profinite persistence)**.** *Let $F(t) = \sum a_n t^n \in \mathbf{Q}[[t]]$ and let $h(a_n)$ denote logarithmic Weil height. If*

$$h(a_n) = o(n) \qquad \text{but} \qquad h(a_n) \neq O((\log n)^2),$$

then, in every fixed base, the total precision profile of every fixed linear box and every fixed blocker box is finite; explicitly, the sums of $\epsilon_p \log p$ and $\eta_p \log p$ over all good primes are finite.

Proof. Infinite total precision in a linear box would make F Mahler by Theorem 3.3; in a blocker box it would do so by Theorem 4.5. The height-gap theorem of Adamczewski, Bell and Smertnig [1, Theorem 6.1] says that a rational-coefficient Mahler series of sublinear coefficient height has height $O((\log n)^2)$, a contradiction. □

The finer height ladder in [1] gives analogous obstructions at the successive thresholds $\log^2 n$, $\log n$, $\log \log n$ and bounded height. These conclusions do not require D-finiteness.

5. D-finite fixed-box escape

We now assume that F is D-finite over $\mathbf{Q}(t)$. Recall that a D-finite k-Mahler series in characteristic zero is rational [5].

Theorem 5.1 (D-finite Witt–Mahler dichotomy)**.** *Let $F \in \mathbf{Q}[[t]] \cap \mathbf{Z}_p[[t]]$ be D-finite and fix $k \geq 2$. Then the following are equivalent.*

(i) *F is rational.*
(ii) *Some fixed order–degree box admits primitive Mahler relations modulo p^m for arbitrarily large m.*
(iii) *Some fixed pair (M, D) admits primitive polynomial blockers modulo p^m for arbitrarily large m.*

Proof. The implications (ii)⇒(i) and (iii)⇒(i) follow from Theorems 2.2 and 4.3, respectively, from Lemma 2.1 when needed, and then from the D-finite–Mahler rationality theorem.

For the converse, the case $F = 0$ is immediate: take the relation $F(t) = 0$ and the blocker $Q = 1$. If $F = A/B \neq 0$ is rational, then after a rational rescaling we may take $A, B \in \mathbf{Z}_p[t]$ with B primitive. Then

$$A(t^k)B(t)F(t) - A(t)B(t^k)F(t^k) = 0,$$

which gives (ii) at every depth after primitive normalization. Taking $Q = B$ makes $QF = A$ a polynomial. Its iterated Cartier sections span a finite-dimensional space, so a fixed (M, D) gives (iii). □

Define the fixed-base linear complexity at depth m by

$$\mu_{k,p}(F; m) = \min\{(q+1)(h+1) : \mathcal{V}_{q,h}(F) \text{ has a primitive relation mod } p^m\}, \tag{14}$$

with value $+\infty$ when there is none.

Corollary 5.2 (Cofinal escape)**.** *If $F \in \mathbf{Q}[[t]] \cap \mathbf{Z}_p[[t]]$ is D-finite and nonrational, then*

$$\mu_{k,p}(F; m) \longrightarrow +\infty.$$

Moreover, for every fixed (M, D) there is a depth after which no primitive (M, D) blocker exists.

Proof. Relations at depth $m + 1$ reduce to primitive relations at depth m, so the minimum in (14) is nondecreasing. If it remained bounded along unbounded depths, one of finitely many boxes would recur arbitrarily deeply, contradicting Theorem 5.1. The blocker assertion is the same contrapositive. □

This corollary asserts escape from every fixed box; it supplies no rate and no lower bound for the order of a P-recursive recurrence.

6. Automatic state complexity modulo p^m

For a sequence $a = (a_n)_{n\geq 0}$ with values modulo p^m, its p-kernel is

$$\mathrm{Ker}_p(a) = \{(a_{p^e n+r})_{n\geq 0} : e \geq 0,\ 0 \leq r < p^e\}.$$

When this set is finite, its cardinality is the state count of the normalized minimal least-significant-digit-first automaton unaffected by leading zeros [6, 9].

Theorem 6.1 (p-kernel escape)**.** *Let $F(t) = \sum a_n t^n \in \mathbf{Q}[[t]] \cap \mathbf{Z}_p[[t]]$ be D-finite and nonrational. Put, in the extended positive integers,*

$$N_m = |\,\mathrm{Ker}_p(a \bmod p^m)| \in \mathbf{N} \cup \{+\infty\}.$$

Then $N_m \to \infty$.

Proof. Reduction modulo p^m maps the kernel at depth $m+1$ onto the kernel at depth m, so N_m is nondecreasing. If $N_m \leq B$ for all m, the exact p-kernel of the $\mathbf{Z}_p$-valued sequence has at most B elements. Indeed, $B + 1$ distinct exact kernel sequences can be separated at finitely many coefficient positions; at a sufficiently large depth all those differences remain visible, contradicting $N_m \leq B$.

Every exact kernel sequence still has rational entries. If the exact p-kernel is finite, its $\mathbf{Q}$-span is finite-dimensional and stable under all Cartier sections. Thus the series is p-regular over $\mathbf{Q}$, and Becker's theorem gives a p-Mahler

equation over $\mathbf{Q}$. Lemma 2.1 puts it in standard form, and D-finite–Mahler rationality contradicts the hypothesis. □

No automaticity assumption is needed for the extended-cardinality conclusion. To read it as an automaton statement one does need every N_m to be finite. This is available for algebraic p-adic series under the prime-power automaticity results summarized in [7], and for rational diagonals under the integral hypotheses of Rowland–Yassawi [8, 9]. It is not known for arbitrary D-finite integer series.

Corollary 6.2 (Automatic state escape)**.** *Under the hypotheses of Theorem 6.1, assume additionally that $a \bmod p^m$ is p-automatic for every m. Then the normalized minimal automaton state counts tend to infinity. In particular this applies to nonrational algebraic series and to nonrational D-finite rational diagonals whenever the cited all-depth automaticity hypotheses hold.*

For a series already known to be integral and algebraic, the conclusion is especially transparent: finite automata exist at each finite Witt level, but no uniform finite state space can survive the inverse tower. The result is complementary to the explicit upper bounds in [9].

7. Changing to the residue-characteristic base

Fixed-base uniformity cannot be replaced by choosing base p at prime p. Nevertheless bounded coefficient degree across infinitely many primes is so restrictive that it forces a stronger collapse.

Proposition 7.1 (Expanding-prefix recurrence lifting)**.** *Let $F = \sum a_n t^n \in \mathbf{Q}[[t]]$. Suppose there are an integer $h \geq 0$ and infinitely many good primes p for which a nonzero vector $(c_{p,0}, \dots, c_{p,h}) \in \mathbf{F}_p^{h+1}$ satisfies*

$$\sum_{j=0}^{h} c_{p,j}\, \overline{a}_{n-j} = 0 \qquad (h < n < p). \tag{15}$$

Then F is rational.

Proof. Form the infinite rational Toeplitz matrix whose row indexed by $n > h$ is

$$(a_n, a_{n-1}, \dots, a_{n-h}).$$

If its columns had full rank, a fixed nonzero rational minor would remain nonzero modulo all but finitely many primes. For every sufficiently large prime in the assumed set, all rows of that minor occur in the range of (15), a contradiction. Thus the columns are dependent over $\mathbf{Q}$, giving an exact constant-coefficient recurrence and hence a rational generating function. □

For a good prime p, let $L_p(F)$ be the least $h \in \{0, \dots, p-2\}$ for which some $0 \neq (c_0, \dots, c_h) \in \mathbf{F}_p^{h+1}$ satisfies

$$\sum_{j=0}^{h} c_j \overline{a}_{n-j} = 0 \qquad (h < n < p).$$

Set $L_p(F) = p-1$ if no such h exists. Proposition 7.1 immediately gives

$$F \notin \mathbf{Q}(t) \quad \Longrightarrow \quad L_p(F) \longrightarrow \infty \quad \text{along the good primes.} \tag{16}$$

Theorem 7.2 (Own-characteristic collapse)**.** *Let $F \in \mathbf{Q}[[t]]$. Suppose that there is an integer $h \geq 0$ and infinitely many primes p of good reduction for which*

$$\sum_{e=0}^{q_p} P_{p,e}(t)\, \overline{F}_p(t^{p^e}) = 0, \qquad q_p \geq 1, \tag{17}$$

where $P_{p,e} \in \mathbf{F}_p[t]$, $\deg P_{p,e} \leq h$, and $P_{p,0} P_{p,q_p} \neq 0$. Then F is constant.

Proof. For $h < n < p$, every term with $e \geq 1$ is invisible in the coefficient of t^n: after multiplication by a polynomial of degree at most h, the nonconstant exponents begin at p. Thus the coefficients of $P_{p,0}$ give a nonzero constant-coefficient recurrence of width at most $h+1$ for an initial segment whose length tends to infinity with p. Proposition 7.1 makes F rational.

Write $F = A/B$ in lowest terms. After discarding finitely many primes, degrees and coprimality are preserved. In characteristic p, $\overline{F}_p(t^{p^e}) = \overline{F}_p(t)^{p^e}$. If F has a pole of order r, the last term of (17) has pole order at least $rp^{q_p} - h$, while all earlier terms have pole order at most $rp^{q_p - 1}$. For $p - 1 > h$ the last term has a unique highest pole order and cannot cancel. Thus B is constant. If A is nonconstant of degree s, the same comparison at infinity gives unique top degree sp^{q_p} versus at most $sp^{q_p-1} + h$. Therefore A is constant. □

Both nonzero endpoint conditions in (17) are necessary for the argument. The theorem closes a particular changing-base route; it does not assert a general cross-prime Mahler rigidity principle.

For a good prime p, define $H_p(F)$ as the minimum of $\max_{e : P_e \neq 0} \deg P_e$ over all $q \geq 1$ and polynomials $P_e \in \mathbf{F}_p[t]$ with $P_0 P_q \neq 0$ satisfying, in $\mathbf{F}_p[[t]]$,

$$\sum_{e=0}^{q} P_e(t) \overline{F}_p(t^{p^e}) = 0.$$

Set $H_p(F) = +\infty$ if no such relation exists. Then Theorem 7.2 has the useful contrapositive

$$F \text{ nonconstant} \quad \Longrightarrow \quad H_p(F) \longrightarrow +\infty \quad \text{along the good primes.} \tag{18}$$

8. Sharpness and counterexamples

8.1. **Full own-characteristic rank one, with an exact degree barrier.** Let

$$C(t) = \sum_{n\geq 0} \binom{2n}{n} t^n = (1-4t)^{-1/2}. \tag{19}$$

For odd p, Lucas' theorem gives

$$\binom{2(pn+r)}{pn+r} \equiv \begin{cases} \binom{2r}{r}\binom{2n}{n}, & 0 \leq r \leq (p-1)/2, \\ 0, & (p+1)/2 \leq r < p \end{cases} \pmod{p}.$$

Thus every own-p Cartier section of C mod p is a scalar multiple of C or zero, and iteration gives

$$\dim_{\mathbf{F}_p} \operatorname{span}\{\Lambda_w C : w \text{ an own-}p \text{ word}\} = 1. \tag{20}$$

At $p = 2$, $C \bmod 2 = 1$, so (20) holds for every prime, not merely at the first Cartier level.

There is nevertheless an exact coefficient-degree cost. For odd p, any endpoint-nonzero relation

$$\sum_{e=0}^{q} P_e(t) C(t^{p^e}) = 0, \qquad P_e \in \mathbf{F}_p[t], \quad \deg P_e \leq h, \quad P_0 P_q \neq 0,$$

must satisfy

$$h \geq \frac{p^{q-1}(p-1)}{2}. \tag{21}$$

Indeed, in the quadratic function field $\mathbf{F}_p(t, C)$, put $x = 1 - 4t$. Then

$$C(t^{p^e}) = C(t) x^{-(p^e-1)/2}.$$

For the x-adic valuation, one has $\operatorname{ord}_x(P_q) \leq \deg P_q \leq h$. The pole contributed by the top iterate therefore cannot be cancelled by the preceding iterates unless (21) holds. For $q = 1$ equality is attained by

$$C(t) = (1-4t)^{(p-1)/2} C(t^p) \pmod{p}. \tag{22}$$

Consequently $H_p(C) = (p-1)/2$ for odd p.

Yet C is D-finite and nonrational, so it is not k-Mahler in characteristic zero for any fixed $k \geq 2$ [5]. In particular, rank one in a base varying with p cannot replace the all-depth, fixed-base hypotheses; the necessary degree escapes exactly as predicted by (18). Since C is integral and algebraic, its reductions modulo every p^m are p-automatic under the standard prime-power automaticity theorem [7]. Consequently Corollary 6.2 shows that their normalized minimal automaton state counts tend to infinity.

8.2. **Rational at every finite depth without uniform complexity.** Fix p and set

$$H_p(t) = \sum_{r\geq 1} p^r t^{r^2}, \qquad F_p(t) = \frac{1}{1-2t} + H_p(t). \tag{23}$$

Modulo p^m, the first series is the polynomial obtained by retaining $1 \leq r < m$; hence $F_p \bmod p^m$ is rational for every m. Nevertheless H_p, and therefore F_p, is non-Mahler in every integer base. Indeed its coefficient height is

$$h([t^n]H_p) = \begin{cases} \sqrt{n}\log p, & n \text{ is a nonzero square,} \\ 0, & \text{otherwise,} \end{cases}$$

which is sublinear but not $O((\log n)^2)$. The height-gap theorem of Adamczewski, Bell and Smertnig [1] rules this out for a rational- coefficient Mahler series. Subtracting the rational summand preserves Mahlerity, so F_p is likewise non-Mahler.

This example shows that "rational modulo every p^m" has no lifting force without a uniform bound on the rational, Mahler or Cartier complexity.

9. An adelic one-pole reversal

We finish with an analytic–arithmetic consequence of having recognized one fixed-base Mahler equation. It is a short assembly of the canonical Mahler denominator and local pole theorem of Adamczewski–Bell–Smertnig, not a new Mahler-denominator theorem.

Theorem 9.1 (One–Mahler adelic pole reversal)**.** *Let*

$$F(t) = \sum_{n\geq 0} a_n t^n \in \mathbf{Q}[[t]]$$

be k-Mahler for some fixed $k \geq 2$. For every rational prime p, let $R_p(F)$ be its p-adic Taylor radius and assume $R_p(F) \geq 1$. Suppose the complex germ of F extends meromorphically to $\mathbb{D} = \{t \in \mathbf{C} : |t| < 1\}$ and that its set of poles in $\mathbb{D}$ is exactly $\{\rho\}$, where $0 < \rho < 1$. Then

$$\tau = \rho^{-1}$$

is a Pisot number or a Salem number.

Proof. Let $d(t)$ be the normalized k-Mahler denominator of F in the sense of [1, Definition 3.9]. Since F has rational coefficients, its defining denominator ideal is Galois invariant; uniqueness of the normalization gives $d \in \mathbf{Q}[t]$. By definition, for some $r \geq 1$ and $q_i(t) \in \overline{\mathbf{Q}}[t]$,

$$d(t)F(t) = \sum_{i=1}^{r} q_i(t)F(t^{k^i}). \tag{24}$$

Both sides are meromorphic on $\mathbb{D}$, so the identity theorem extends (24) throughout $\mathbb{D}$. If $d(\rho) \neq 0$, every term on the right is holomorphic at $t = \rho$: indeed $0 < \rho^{k^i} < \rho$, and ρ is the only pole of F in the disk. Equation (24)

would then make F holomorphic at ρ, a contradiction. Thus $d(\rho) = 0$ and ρ is algebraic.

Let v be a nonarchimedean absolute value of $\overline{\mathbf{Q}}$. If $|\rho|_v < 1$, Proposition 6.4 of [1] forces the v-adic Taylor radius of F to be strictly smaller than 1, contrary to the hypothesis. For rational coefficients this is the radius at the underlying rational prime. Hence $|\rho|_v \geq 1$ at every finite place, equivalently $|\tau|_v \leq 1$, so τ is an algebraic integer.

Let $\rho' \neq \rho$ be any algebraic conjugate of ρ under an embedding into $\mathbf{C}$. Since $d \in \mathbf{Q}[t]$, one has $d(\rho') = 0$. The proof of [1, Proposition 6.4] gives the stronger local conclusion that a nonzero denominator root strictly inside the complex unit disk is an actual pole of the meromorphic continuation. Thus $|\rho'| < 1$ would produce a second pole in $\mathbb{D}$. Consequently every conjugate $\tau' \neq \tau$ satisfies $|\tau'| \leq 1$.

If all these inequalities are strict, τ is Pisot. Otherwise let $P \in \mathbf{Z}[x]$ be the monic irreducible polynomial of τ and choose a conjugate β on the unit circle. Then $\overline{\beta} = \beta^{-1}$ is also a root of P, so P and its reciprocal polynomial have a common root. Irreducibility makes P reciprocal up to sign. Its roots are closed under inversion; since τ is the only root outside the unit disk, the only roots off the unit circle are τ and τ^{-1}. Thus τ is Salem. □

Corollary 9.2 (Profinite pole interface)**.** *Assume the analytic and finite-place hypotheses of Theorem 9.1. It is enough to replace the assumption that F is already Mahler by either of the following, in one fixed base:*

(i) *a fixed linear box has divergent total precision as in Theorem 3.3;*
(ii) *a fixed blocker box has divergent total precision as in Theorem 4.5.*

Under either condition, ρ^{-1} is Pisot or Salem.

Proof. In case (i), Theorem 3.3 and Lemma 2.1 make F Mahler in the fixed base. In case (ii), Theorem 4.5 does so directly. Theorem 9.1 then applies. □

10. Limits and further questions

The results separate six levels that should not be conflated:

hypothesis	conclusion
fixed Mahler box, divergent total local precision	exact k-Mahler relation
bounded blocker box, divergent total local precision	exact k-Mahler relation, with no effective output box
either criterion in two independent bases	rationality, without D-finiteness
D-finite plus either one-base criterion	rationality
D-finite and nonrational	reduced p-kernel cardinalities tend to infinity
one recognized base plus the adelic unique-pole hypotheses	reciprocal pole is Pisot or Salem

The present methods do not give an escape rate, a finite positive certificate for rationality, or automaticity of reductions for all D-finite series. The pole reversal does not construct a persistent Mahler window, prove uniqueness of an inner pole, or handle a boundary point tending to 1; in particular it does not prove the Salem–Boyd closure conjecture. A natural next problem is to bound the stabilized Smith exponent in terms of arithmetic data of a D-finite differential equation. Another is to make the blocker theorem effective by bounding the Mahler equation produced from (k, M, D). Both require input absent from the compactness and Nullstellensatz proofs here.

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